DATA MINING ASSIGNMENT REPORT

Credit Card Fraud Detection using Data Mining Techniques

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TABLE OF CONTENTS

1. EXECUTIVE SUMMARY

This project focuses on the application of data mining techniques for detecting fraudulent financial transactions using a real-world transaction dataset. Fraud detection is one of the most important applications of data mining in the banking and financial sector because fraudulent activities can lead to major financial losses and security risks for organizations and customers.

The primary objective of this project is to analyse transaction-related data and identify patterns associated with fraudulent activities using multiple data mining techniques. The project demonstrates the complete data mining lifecycle, including data acquisition, preprocessing, exploratory data analysis, classification, clustering, association rule mining, and performance evaluation.

The dataset used in this project contains 32,300 transaction records with multiple attributes such as transaction amount, account balance, merchant risk score, transaction timing, and fraud indicators. The dataset was analysed to identify hidden relationships and suspicious transaction behaviours.

Various preprocessing techniques such as missing value handling, duplicate removal, normalization, and exploratory data analysis were performed to improve the quality of the dataset before model implementation.

Multiple data mining algorithms were implemented in this project. Decision Tree and Naive Bayes algorithms were used for classification tasks. K-Means clustering was implemented for identifying transaction behaviour patterns, while the Apriori algorithm was used for association rule mining.

The performance of the classification models was evaluated using accuracy, precision, recall, and F1-score metrics. The results showed that the Decision Tree classifier achieved balanced fraud detection performance, while the Naive Bayes classifier achieved higher precision but lower recall.

This project successfully demonstrates how data mining techniques can be applied in real-world fraud detection systems to improve financial security and decision-making processes.

2. INTRODUCTION TO DATA MINING

Definition of Data Mining

Data mining is the process of extracting useful patterns, hidden relationships, trends, and valuable knowledge from large datasets using statistical, computational, and machine learning techniques. It helps organizations make better decisions by discovering meaningful insights from raw data.

Data mining is widely used in industries such as banking, healthcare, marketing, education, cybersecurity, and e-commerce to analyse large amounts of data efficiently.

Relationship Between Data Mining and KDD

Knowledge Discovery in Databases (KDD) is the complete process of discovering useful knowledge from data. Data mining is one of the major steps in the KDD process.

The KDD process includes the following stages:

1. Data Selection
   Relevant data is selected from various sources.
2. Data Cleaning
   Missing values, inconsistencies, and duplicate records are removed.
3. Data Transformation
   Data is transformed into a suitable format for analysis.
4. Data Mining
   Algorithms are applied to identify patterns and relationships.
5. Pattern Evaluation
   Useful patterns are identified and evaluated.
6. Knowledge Representation
   The discovered knowledge is presented in understandable formats such as graphs and reports.

Difference Between Data Mining, Statistics, and Machine Learning

Concept	Description
Data Mining	Extracts hidden patterns and useful knowledge from large datasets
Statistics	Focuses on mathematical analysis, probability, and data interpretation
Machine Learning	Builds models that automatically learn from data and improve predictions

Real World Applications of Data Mining

1. Fraud Detection

Banks and financial institutions use data mining techniques to identify suspicious transactions and detect fraudulent activities.

2. Recommendation Systems

E-commerce platforms use recommendation systems to suggest products based on customer behaviour and purchase history.

3. Healthcare Prediction

Healthcare organizations use data mining to predict diseases and analyze patient records.

4. Customer Segmentation

Businesses use data mining to divide customers into groups based on purchasing behaviour.

5. Spam Email Detection

Email systems use data mining algorithms to classify spam and non-spam emails.

3. DATASET DESCRIPTION

Dataset Information

The dataset used in this project is a financial transaction dataset related to fraud detection.

Dataset Name

credit_fraud.csv.xlsx

Total Number of Records

32,300 rows

Total Number of Attributes

10 columns

Attribute Description

Attribute	**Description**
age	Age of the customer
transaction_amount	Amount involved in the transaction
account_balance	Current account balance of customer
num_transactions_today	Number of transactions made in a day
is_foreign_transaction	Indicates whether transaction is foreign
transaction_hour	Time at which transaction occurred
prev_fraud_flag	Indicates previous fraud history
merchant_distance_km	Distance of merchant location
merchant_risk_score	Risk level associated with merchant
is_fraud	Target variable indicating fraud or non-fraud

Dataset Objective

The main objective of this dataset is to identify whether a transaction is fraudulent or legitimate based on transaction behaviour and related attributes.

4. DATA PREPROCESSING AND EXPLORATION

Missing Value Handling

The dataset contained missing values in several columns including age, transaction amount, account balance, merchant risk score, and transaction hour.

Mean imputation technique was used to replace missing numerical values to maintain dataset consistency.

Missing Values Output

None

	transaction_amount	account_balance	num_transactions_today	...	merchant_distance_km	merchant_risk_score	is_fraud
count	31653.000000	31655.000000	31654.000000	...	31657.000000	31656.000000	32300.000000
mean	7471.580234	25020.772509	24.924591	...	2500.266033	5.512513	0.481734
std	4324.127410	14423.748726	14.150679	...	1444.739723	2.594919	0.499674
min	1.070000	-99999.000000	1.000000	...	0.100000	1.000000	0.000000
25%	3723.400000	12510.295000	13.000000	...	1250.000000	3.300000	0.000000
50%	7475.070000	25050.490000	25.000000	...	2507.000000	5.500000	0.000000
75%	11199.090000	37530.215000	37.000000	...	3741.300000	7.800000	1.000000
max	14999.710000	49998.900000	49.000000	...	4999.800000	15.000000	1.000000

Duplicate Record Removal

Duplicate records can reduce model accuracy and create biased results.
Therefore, duplicate entries were removed from the dataset.

Dataset Size Before Removing Duplicates

32,300 records

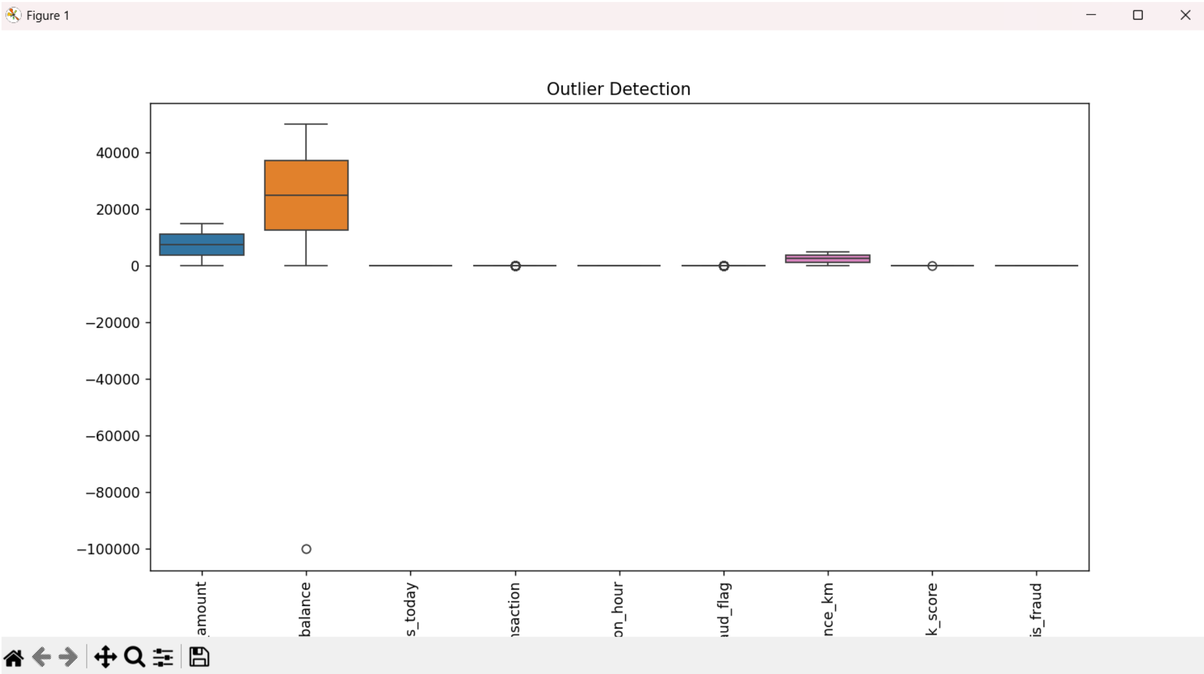
Dataset Size After Removing Duplicates

32,002 records

Outlier Detection

Outliers are extreme values that can negatively affect model performance.
Boxplots were used to identify unusual values in transaction-related attributes.

Outlier Detection Graph



Data Normalization

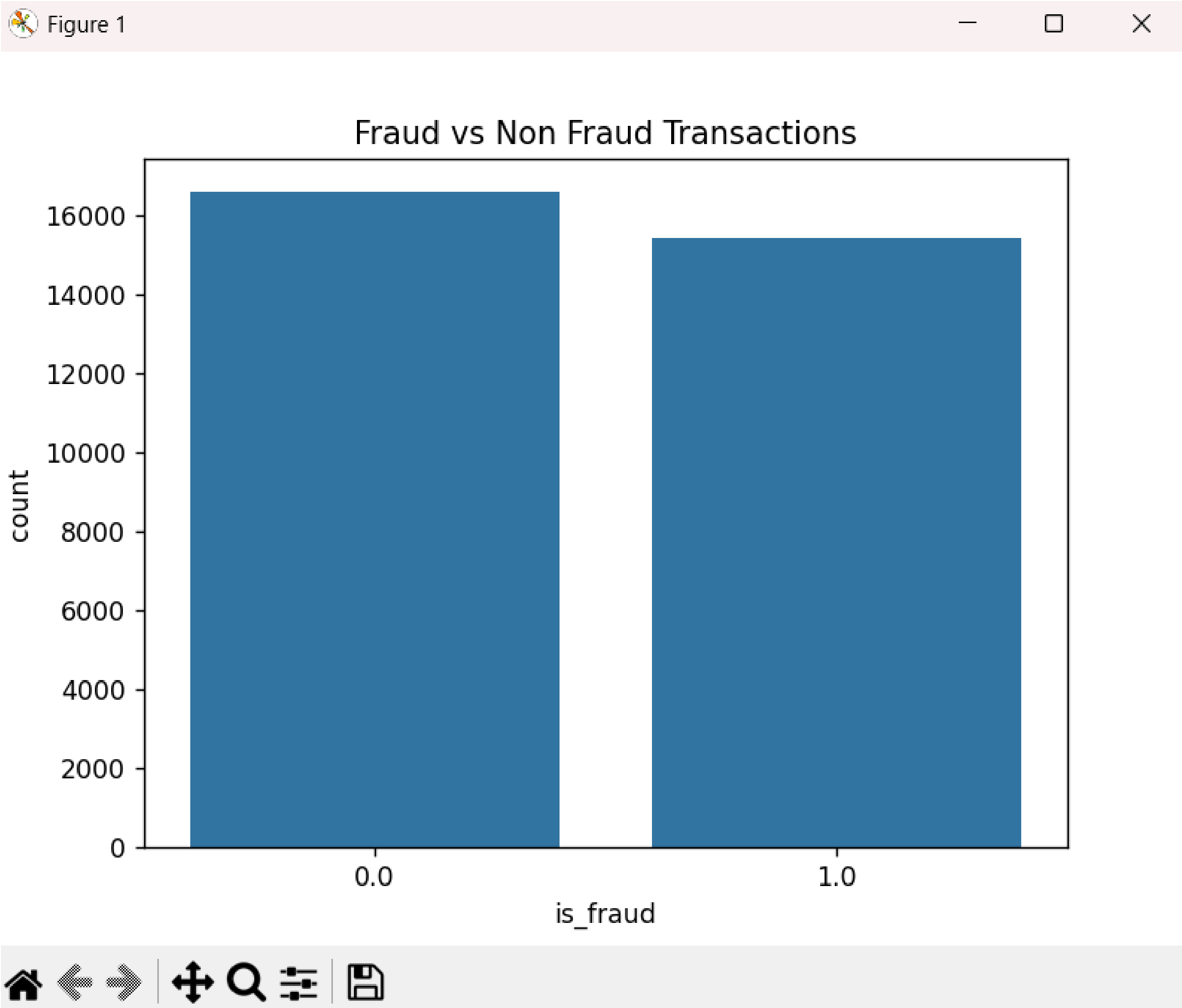
Feature scaling was performed using StandardScaler to ensure that all numerical features were on the same scale before model training.

Normalization helps improve the performance of machine learning algorithms.

Exploratory Data Analysis

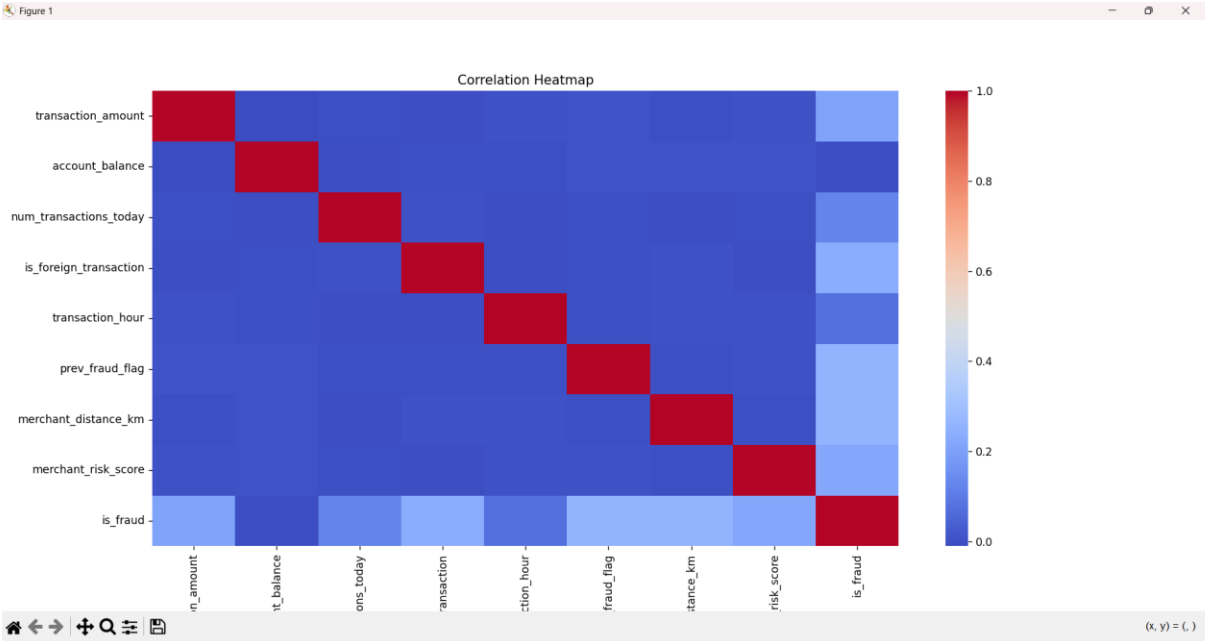
Exploratory Data Analysis was performed to understand the structure and behavior of the dataset.

Fraud vs Non-Fraud Distribution



The graph shows the distribution of fraudulent and legitimate transactions in the dataset.

Correlation Heatmap



The heatmap was used to identify relationships between numerical attributes in the dataset.

5. METHODOLOGY

Step 1: Data Collection

The dataset was collected for educational and analytical purposes related to financial fraud detection.

Step 2: Data Cleaning

The following preprocessing techniques were applied:

- Missing value handling
- Duplicate removal
- Outlier analysis
- Data normalization

These preprocessing steps improved the overall quality of the dataset.

Step 3: Feature Selection

The target variable selected for classification was:

is_fraud

The remaining attributes were used as input features for model training.

Step 4: Model Training

Multiple data mining algorithms were implemented for analysis and comparison.

Decision Tree Classifier

Decision Tree was selected because it provides easy interpretation and handles non-linear relationships effectively.

Naive Bayes Classifier

Naive Bayes was selected because it is computationally efficient and performs well on large datasets.

K-Means Clustering

K-Means clustering was used to group transactions based on similar transaction behaviour patterns.

Apriori Algorithm

Apriori algorithm was implemented to identify hidden associations between transaction attributes.

Step 5: Model Evaluation

The models were evaluated using the following performance metrics:

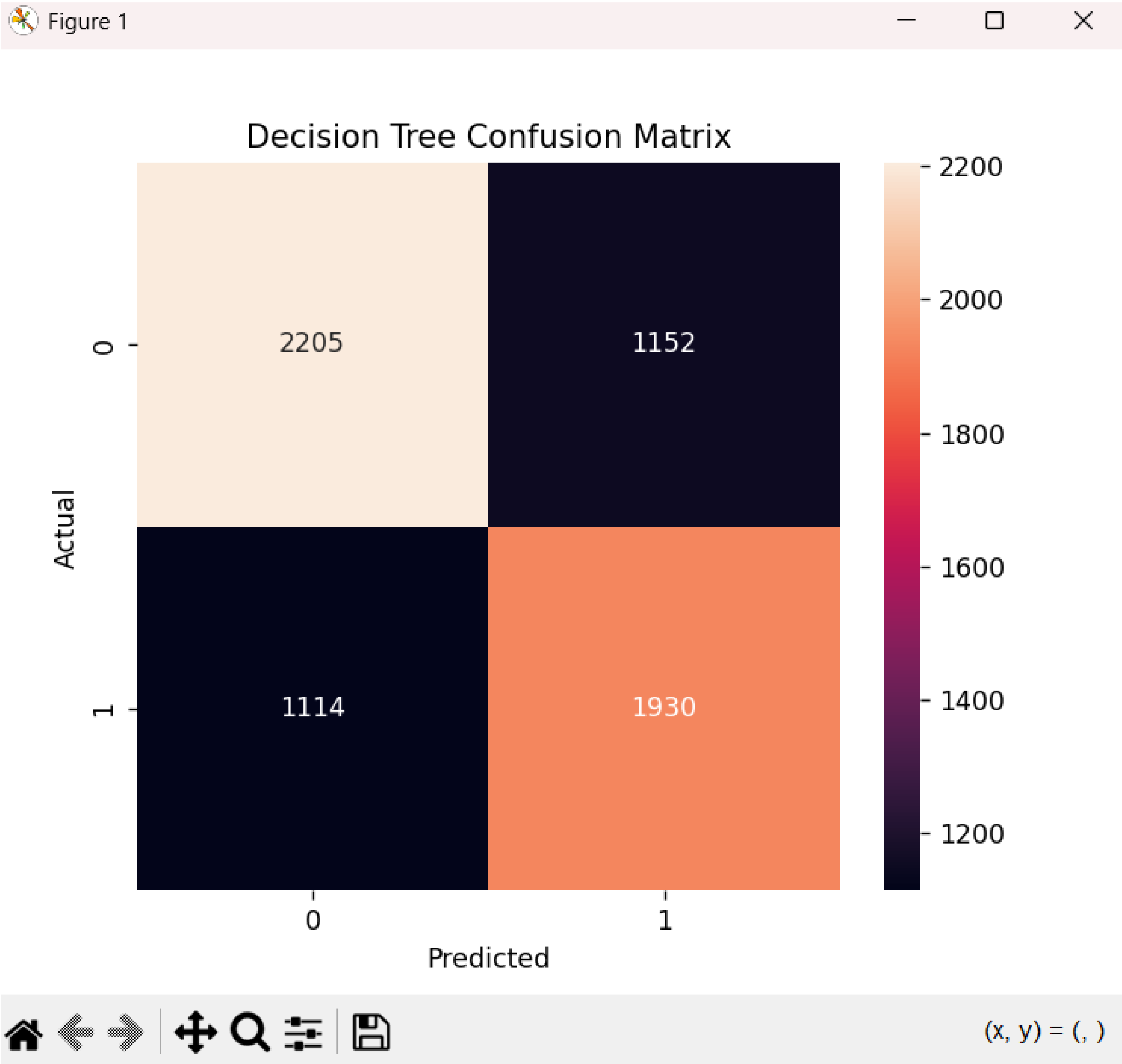
- Accuracy
- Precision
- Recall
- F1 Score
- Silhouette Score

6. DATA MINING TECHNIQUES IMPLEMENTATION

Decision Tree Classification

Decision Tree classification algorithm was implemented to classify fraudulent and legitimate transactions.

Decision Tree Confusion Matrix



Decision Tree Performance

- Accuracy: 64.60%
- Precision: 62.62%

- Recall: 63.40%
- F1 Score: 63.01%

Interpretation

The Decision Tree classifier achieved balanced performance in detecting fraudulent transactions. The model successfully identified fraudulent transaction patterns while maintaining balanced precision and recall values.

Naive Bayes Classification

Naive Bayes classification algorithm was implemented for fraud prediction.

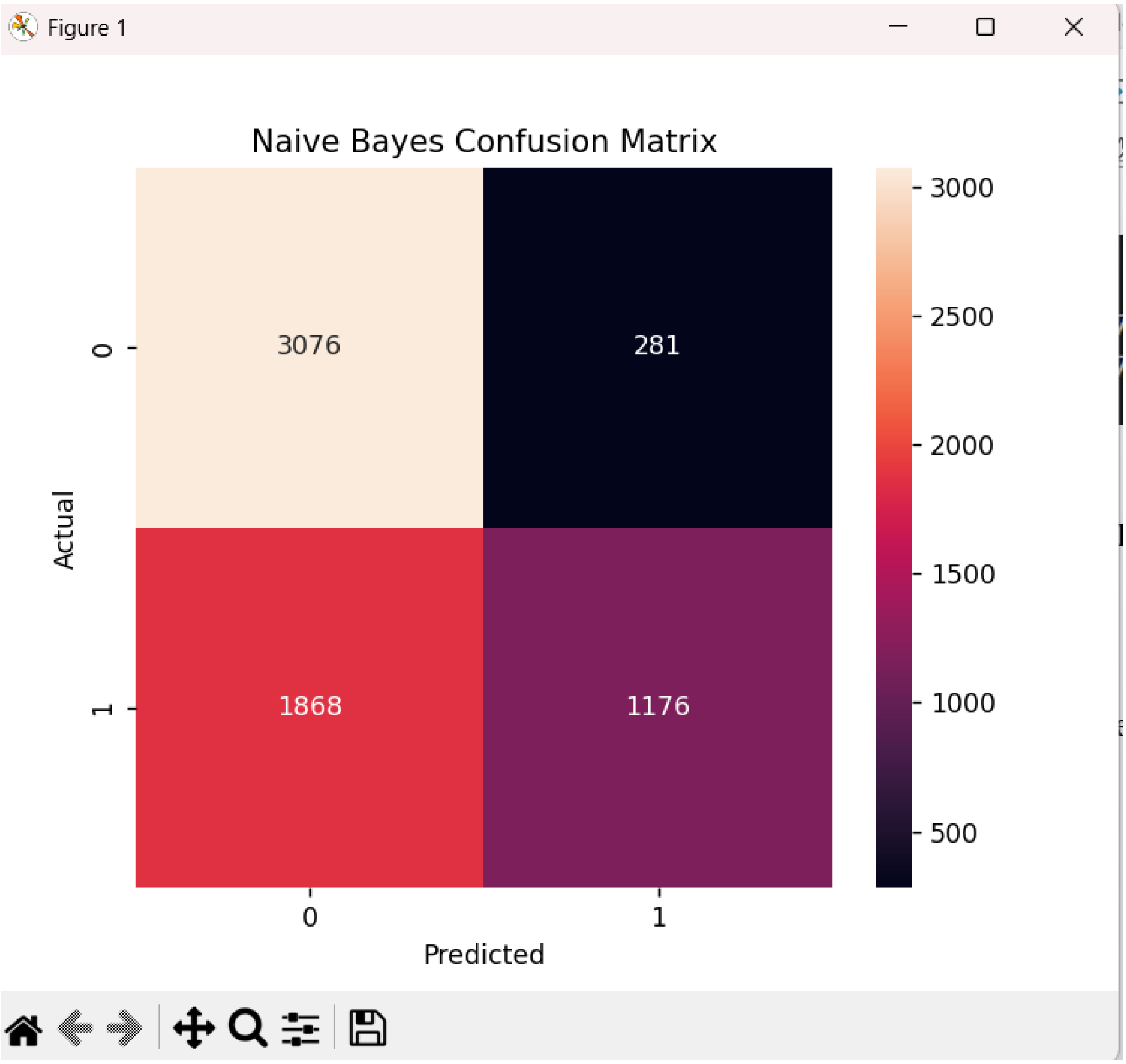
Naive Bayes Code Output

```
Naive Bayes Results
Accuracy: 0.6642712076238088
Precision: 0.8071379547014413
Recall: 0.38633377135348224
F1 Score: 0.5225505443234837
              precision    recall  f1-score   support

         0.0       0.62      0.92      0.74      3357
         1.0       0.81      0.39      0.52      3044

    accuracy                           0.66      6401
   macro avg       0.71      0.65      0.63      6401
weighted avg       0.71      0.66      0.64      6401
```

Naive Bayes Confusion Matrix



Naive Bayes Performance

- Accuracy: 66.43%
- Precision: 80.71%
- Recall: 38.63%
- F1 Score: 52.25%

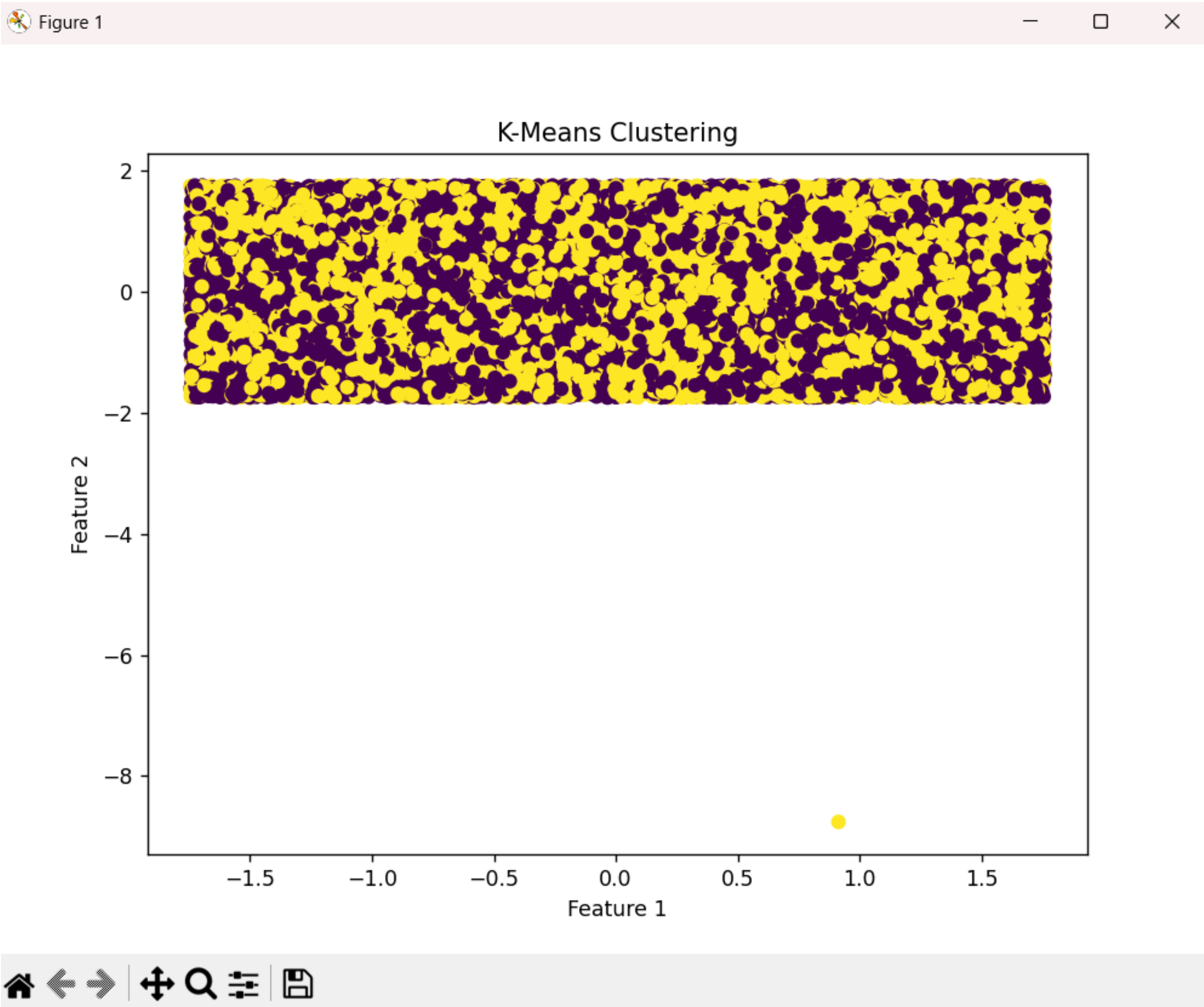
Interpretation

Naive Bayes achieved high precision but lower recall. This indicates that the model missed several fraudulent transactions despite achieving good precision.

K-Means Clustering

K-Means clustering algorithm was used to identify groups of transactions with similar patterns.

K-Means Clustering Graph



Clustering Analysis

The clustering algorithm grouped transactions based on transaction behaviour and similarity patterns.

Association Rule Mining

Association rule mining was implemented using the Apriori algorithm.

Association Rules Output

Association Rules

	antecedents	consequents	support	confidence	lift
0	frozenset({transaction_amount})	frozenset({account_balance})	0.488001	0.999936	0.999967
1	frozenset({transaction_amount})	frozenset({merchant_distance_km})	0.249016	0.510245	0.997303
2	frozenset({num_transactions_today})	frozenset({account_balance})	0.475220	1.000000	1.000031
3	frozenset({is_foreign_transaction})	frozenset({account_balance})	0.142866	1.000000	1.000031
4	frozenset({transaction_hour})	frozenset({account_balance})	0.367040	1.000000	1.000031
5	frozenset({merchant_distance_km})	frozenset({account_balance})	0.511593	0.999939	0.999970
6	frozenset({account_balance})	frozenset({merchant_distance_km})	0.511593	0.511609	0.999970
7	frozenset({merchant_risk_score})	frozenset({account_balance})	0.213362	1.000000	1.000031
8	frozenset({num_transactions_today})	frozenset({merchant_distance_km})	0.241610	0.508417	0.993730
9	frozenset({transaction_hour})	frozenset({merchant_distance_km})	0.188457	0.513451	1.003571
10	frozenset({merchant_risk_score})	frozenset({merchant_distance_km})	0.108493	0.508494	0.993883
11	frozenset({transaction_amount, num_transaction...	frozenset({account_balance})	0.231954	1.000000	1.000031
12	frozenset({transaction_amount, transaction_hour})	frozenset({account_balance})	0.179145	1.000000	1.000031
13	frozenset({merchant_distance_km, transaction_a...	frozenset({account_balance})	0.248984	0.999875	0.999906
14	frozenset({transaction_amount, account_balance})	frozenset({merchant_distance_km})	0.248984	0.510213	0.997242
15	frozenset({transaction_amount})	frozenset({merchant_distance_km, account_balan...	0.248984	0.510181	0.997239
16	frozenset({transaction_amount, merchant_risk_s...	frozenset({account_balance})	0.104837	1.000000	1.000031
17	frozenset({transaction_amount, num_transaction...	frozenset({merchant_distance_km})	0.116743	0.503301	0.983731
18	frozenset({num_transactions_today, transaction...	frozenset({account_balance})	0.174145	1.000000	1.000031
19	frozenset({merchant_distance_km, num_transacti...	frozenset({account_balance})	0.241610	1.000000	1.000031
20	frozenset({num_transactions_today, account_bal...	frozenset({merchant_distance_km})	0.241610	0.508417	0.993730
21	frozenset({num_transactions_today})	frozenset({merchant_distance_km, account_balan...	0.241610	0.508417	0.993791
22	frozenset({num_transactions_today, merchant_ri...	frozenset({account_balance})	0.102119	1.000000	1.000031
23	frozenset({merchant_distance_km, transaction_h...	frozenset({account_balance})	0.188457	1.000000	1.000031
24	frozenset({transaction_hour, account_balance})	frozenset({merchant_distance_km})	0.188457	0.513451	1.003571
25	frozenset({transaction_hour})	frozenset({merchant_distance_km, account_balan...	0.188457	0.513451	1.003633
26	frozenset({merchant_distance_km, merchant_risk...	frozenset({account_balance})	0.108493	1.000000	1.000031
27	frozenset({merchant_risk_score, account_balance})	frozenset({merchant_distance_km})	0.108493	0.508494	0.993883
28	frozenset({merchant_risk_score})	frozenset({merchant_distance_km, account_balan...	0.108493	0.508494	0.993943
29	frozenset({merchant_distance_km, transaction_a...	frozenset({account_balance})	0.116743	1.000000	1.000031
30	frozenset({transaction_amount, num_transaction...	frozenset({merchant_distance_km})	0.116743	0.503301	0.983731
31	frozenset({transaction_amount, num_transaction...	frozenset({merchant_distance_km, account_balan...	0.116743	0.503301	0.983791

Association Rule Analysis

The Apriori algorithm identified hidden relationships between transaction-related attributes and fraud behaviour.

7. RESULTS AND ANALYSIS

Decision Tree Results Analysis

The Decision Tree classifier provided balanced fraud detection performance with moderate accuracy and recall values.

The model was able to identify fraudulent transactions more effectively compared to Naive Bayes.

Naive Bayes Results Analysis

Naive Bayes achieved slightly higher accuracy and precision. However, the recall value was significantly lower.

This means that the model failed to identify many fraudulent transactions.

Business Implications

Fraud detection systems require high recall because missing fraudulent transactions can result in severe financial losses.

Although Naive Bayes achieved higher precision, Decision Tree was more effective for fraud detection due to its balanced performance.

8. MODEL COMPARISON

Metric	**Decision Tree**	**Naive Bayes**
Accuracy	64.60%	66.43%
Precision	62.62%	80.71%
Recall	63.40%	38.63%
F1 Score	63.01%	52.25%

Comparative Analysis

The Decision Tree model performed better overall because it achieved significantly higher recall.

In fraud detection systems, recall is extremely important because identifying fraudulent transactions is more critical than achieving slightly higher accuracy.

9. CONCLUSION

This project successfully demonstrated the use of data mining techniques for detecting fraudulent financial transactions.

Several preprocessing techniques such as missing value handling, duplicate removal, normalization, and exploratory data analysis were performed before implementing machine learning models.

Decision Tree and Naive Bayes classification algorithms were implemented and evaluated. K-Means clustering and association rule mining techniques were also applied to identify hidden transaction patterns.

The Decision Tree classifier provided better overall fraud detection performance due to its balanced precision and recall values.

The project demonstrates the importance of data mining in improving fraud detection systems and financial security.

10. RECOMMENDATIONS

1. Use advanced ensemble learning techniques such as Random Forest and XGBoost for improved fraud detection performance.
2. Implement real-time fraud monitoring systems for faster fraud detection.
3. Improve feature engineering techniques for better model accuracy.
4. Use anomaly detection algorithms for identifying unknown fraud patterns.

11. FUTURE SCOPE

1. Deep learning models can be implemented for more advanced fraud detection systems.
2. Real-time streaming analytics can be used for continuous fraud monitoring.
3. Cloud computing platforms such as AWS and Google Cloud can be used for large-scale fraud analysis.
4. Big data technologies such as Apache Spark can improve processing efficiency for large datasets.

12. REFERENCES

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5. Python Official Documentation